

CUTTING EDGE

With technology taking away much of the blood-and-gut messiness of surgery, modern operating rooms get closer to the world of *Star Trek*. For the patient this means speedy recovery and less pain but also a much heavier medical bill.

■ By Damayanti Datta

The operation taking place in the green tiled operating room at the All India Institute of Medical Sciences (AIIMS) is almost eerie. The patient, anaesthetised and swathed in blue drapes, is lying face up on a narrow table. But no one is touching her. Instead, the lead surgeon, Dr Arvind Kumar, sits some distance away, bowed over what looks like a video-game console. Yet action is taking place. As Kumar's thumbs and forefingers hit the controls, a four-armed robot springs to life. Nurses, technicians, doctors stand motionless, staring at a sea of monitors, while tiny metal "fingers"—inserted through pencil-sized holes in the patient's chest—carry out Kumar's commands: clamping, cutting, sewing.

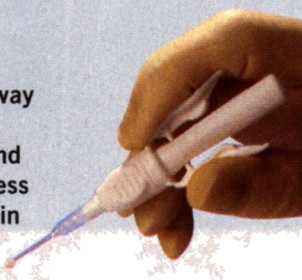
The AIIMS robot may not intone, "This won't

hurt a bit", like C3PO of *Star Trek*, but surgery in the new millennium has gone through a galactic change. Technology is now the driving force, minimally-invasive the gold-standard and precision the mantra. Doctors mindmeld with machines, target drugs and instruments, don't use hands to

Bleeding was controlled by electricity or ultrasound once. Today electrothermal systems ensure minimum loss.



Sutures and staplers are now giving way to tissue adhesives and soldering. Less scar, less pain for patient.



Thanks to new generation CT, MRI and PET scans, doctors can explore every nook and cranny. Diagnosis much more accurate.

